INCH-POUND

A-A-59543
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SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION

SCREEN, HALFTONE, CONTACT, FOR PHOTOLITHOGRAPHY

The General Services Administration has authorized the use of this Commercial Item Description (CID) for all federal agencies.

1. SCOPE.

1.1 Abstract. This commercial item description covers photographic halftone contact printing screens used in the preparation of halftone negatives for photolithography. The contact printing screen consists of a uniform, photographically prepared grid of vignetted dots on a transport polyester film base. The screen is a magenta positive type.

2. CLASSIFICATION. The screens shall be of the following sizes and rulings, types and classes:

Type I – Size 24 by 24 inches
Class A – Ruling 150 - line
Class B – Ruling 300 - line

Type II – Size 29 by 29 inches
Class A – Ruling 150 - line
Class B – Ruling 300 - line

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: US Army Topographic Engineering Center, 7701 Telegraph Road, ATTN: CEERD-TS-T, Alexandria, VA 22315-3864.

FSC 3610

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3. SALIENT CHARACTERISTICS.

3.1 Screen dimensions. The screens shall conform to the following dimensions.

Field	Type I	Type II	Dimension Tolerance
Working area	24 by 24 inches	29 by 29 inches	+ 0.5, - 0.0 inch
Overall size	25 by 25 inches	31 by 31 inches	+ 0.5, - 0.0 inch
Thickness	0.009 inch	0.009 inch	± 0.001 inch

3.2 Screen base. The film base, emulsion, and protective coating, if used, shall be free from any defects. If a border is provided outside the usable screen area of the film, it shall be transparent , except for identification marking.

3.3 Screen dot structure. The screen pattern shall consist of identical vignetted dots uniformly spaced in rows inclined 45 degrees, ± 3 degrees, to the film margin. The density of each dot shall be graduated from the maximum density at the center to transparent at the interstice between four dots. When exposures are made, the density gradient of the dot formation shall reproduce all steps of a standard gray scale and shall permit extending, contracting, or shifting the reproduced dot size range. The irregularities shall not be larger than 0.10 inch in diameter, and shall not reproduce photographically. In addition the irregularities shall not comprise, (a) the complete absence of a dot, (b) the complete filling in of the area between four dots, and (c) defects which are visible to the unaided eye in a 1:1 photographic reproduction of the screen.

3.4 Screen ruling. The screen ruling shall be defined as the number of dots per lineal inch along the angle of dot formation. The actual dot count shall not vary by more than 5 percent from the commercial rating along any random selected row.

3.5 Flatness. The screen shall not have any areas that may induce wrinkle formation or out-of-contact areas.

3.6 Performance. The screen shall produce a uniform tint pattern over the entire usable screen area. The tint negative shall not have any spots, lines, tonal differences attributed to the screen except those permitted on screen base and those defects inherent in the exposure and processing of the film. The screen shall reproduce or modify the tonal graduation of a density step wedge or equivalent copy.

3.7 Instructions. Instructions for the proper care, handling and cleaning of the screen shall be provided with each container. The screen shall be identified by permanently printing the following information in the margin of the emulsion side of the screen: contact screen, screen side.

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3.8 Filters. Two filters shall be provided with each contact screen. One shall be a Number Four (4) yellow and the other shall be a Number Thirty (30) rose. Each filter shall be two inches square or larger.

4. REGULATORY REQUIREMENTS.

4.1 Recycled, recovered, or environmentally preferable materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. QUALITY ASSURANCE PROVISIONS.

5.1 Market acceptability. The screens offered shall be the latest version of the standard product of the supplier that has been used in the field for at least one year by commercial organizations.

5.2 Product conformance. The screens offered shall meet the salient characteristics of this CID, conform to the manufacture's drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

6. PACKAGING.

6.1 Packaging. Preservation, packing, and marking shall be as specified in the contract or order (see 7.1)

7. NOTES

7.1 Ordering data. Acquisition documents should include:

a) Title, number, and date of this document.
b) Type and class of screen required (see 2.1).
c) Level of preservation and packing required (see 6.1).

7.2 Intended use. The halftone screen is used to reproduce continuous-tone products on offset presses. Since the press can only apply solid ink to the paper, shades of gray or paler colors are created by placing the ink dots in clusters separated by white spaces.

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7.3 Part identification number (PIN). The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.

This example describes a part numbering system for CID A-A-59543

AA59543XX

- Class A = Ruling 150 - line
  Class B = Ruling 300 - line
- Type I = Size 24 by 24 inches
  Type II = Size 29 by 29 inches
- Document identifier

Example: AA59543IIA— A 29 by 29 inch screen, ruling 150 - line

7.4 Sources of documents. Defense Automated Printing Service, Building 4D, NPM-DoDSSP, 700 Robbins Avenue, Philadelphia, PA 19111-5094.

7.5 Subject term (key word) listing.

Photography
Lithography
Film
Graphic arts

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